Sahaj Bhadja

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Summary

Software engineer with 2+ years building **data pipelines**, **ML systems on sensor data**, and **full-stack applications** across cloud infrastructure. Experienced translating user requirements into deployed solutions in fast-paced, customer-facing environments.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*

- Built **ML pipelines processing 100K+ sensor trajectory records** for autonomous driving, implementing data ingestion, feature extraction, and model training workflows using PyTorch, enabling reproducible evaluation and rapid iteration.
- Designed **spatial-temporal encoder** integrating HD-map geometry with vehicle sensor data via multi-head attention, improving closed-loop planning stability by 15% and demonstrating ML applied to real-world sensor problems.

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*

- Built **data integration pipelines** processing 5K+ monthly jobs across SQL databases, cloud storage, and external APIs with validation checks, error handling, and automated recovery, achieving 99%+ production reliability.
- Deployed **Python backend services** on Azure serving 500+ users, reducing latency from **1.2s to 350ms** through query optimization, indexing, and performance tuning while supporting CI/CD-backed production releases.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*

- Developed **full-stack platform with Python/Flask backend and React frontend** serving 150+ instructors, translating user workflow requirements into technical specifications and shipping features handling 10K+ daily requests.
- Integrated **ML models (GPT-4o, Claude) into production workflows** with concurrent batch processing, reducing feedback turnaround by **60%** while maintaining stable API performance under instructor workloads.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*

- Engineered **cross-platform deployment pipelines** with Jenkins CI/CD across Windows, Linux, and macOS, automating build, test, and release workflows while maintaining 20+ test suites with PyTest and Mockito.
- Built **API integrations and ETL workflows** using Python, AWS Glue, and S3, improving response times by **30%** and reducing technical debt through OOP design patterns and maintainable service architecture.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*

- Developed **real-time data synchronization platform** using MERN stack and WebSockets, enabling 2M+ record syncs across distributed locations with conflict resolution and live status updates.
- Deployed **Docker-based services on Azure** with Jenkins automation, cutting release cycles by **50%** through containerized builds, staged deployments, and production monitoring integration.

Skills

Languages: **Python**, **TypeScript**, JavaScript, SQL, C++, Java, Bash
ML/Data: PyTorch, sensor data processing, feature engineering, data pipelines, ETL workflows
Full-Stack: React, Next.js, Flask, FastAPI, Node.js, REST APIs, WebSockets
Infrastructure: AWS (S3, Glue, EC2), Azure, Docker, Kubernetes, Linux, Git, CI/CD
Tools: Git, SSH, command-line workflows, Jenkins, testing automation, Agile/Scrum

Projects

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized data processing API** deployed on AWS ECS/Fargate with CodePipeline automation, implementing validation gates and monitoring for production-grade reliability across deployment environments.

Chat2SQL Code | *Java, TypeScript, PostgreSQL, Docker*

- Built an **AI-powered database workspace** supporting 10+ storage systems (PostgreSQL, MongoDB, Redis, Snowflake), enabling schema exploration and query generation across diverse data backends.

CPU Scheduling Simulator Live | *C++, React, JavaScript*

- Built a **systems-level scheduling simulator** implementing FCFS, SJF, Round Robin, and MLQ algorithms with real-time Gantt visualization, demonstrating OS internals and resource allocation optimization.